Group Theory – Exercise set 6

Exercise 1. (1) Find examples where the map $\exp : \mathfrak{g} \to G$ is not surjective.

(2) Find two Lie group homomorphisms $\Phi, \Psi : G \to H$ which are different but such that the induced Lie algebra homomorphisms $\varphi, \psi : \mathfrak{g} \to \mathfrak{h}$ are equal.

Exercise 2. Consider the representation of the Lie group $G = (\mathbb{C}^*)^2$ one $V = \mathbb{C}^4$ defined by

$$(s_1, s_2) \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} s_1 s_2 v_1 \\ s_1^3 v_2 \\ s_1 s_2^5 v_3 \\ s_2 v_4 \end{pmatrix} \qquad \text{for } (s_1, s_2) \in (\mathbb{C}^*)^2,\ \mathbf{v} \in \mathbb{C}^4.$$

(1) Show it is indeed a Lie group representation.

(2) Determine all the 1-dimensional invariant subspaces of V.

(3) By Schur's Lemma, it follows that they are all the irreducible subspaces of V. Use them to find all the invariant subspaces of V.

(4) Determine the induced Lie algebra representation of $\mathfrak{g} = \mathbb{C}^2$ on V.

Exercise 3. Consider the natural representation of $G = S_n$ on $V = \mathbb{R}^n$, by permutation of coordinates:

$$f \cdot \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} v_{f^{-1}(1)} \\ v_{f^{-1}(2)} \\ \vdots \\ v_{f^{-1}(n)} \end{pmatrix} \qquad \text{for } f \in S_n,\ \mathbf{v} \in \mathbb{R}^n.$$

Show that there are exactly two (non-trivial, irreducible) invariant subspaces.

Exercise 4. Show that the standard representation of $\mathsf{GL}(n; \Bbbk)$ is irreducible.

Exercise 5. Consider the representation of $G = \mathbb{R}$ on $V = \mathbb{C}^2$ defined by $\Pi(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$. Show that V has exactly one invariant subspace of dimension 1. Conclude that V is not irreducible, but it does not decompose as a direct sum of irreducible representations.

Bonus exercise. Show that the Laplacian is rotationally invariant: if $f \in C^2(\mathbb{R}^3)$ and $A \in \mathsf{O}(3)$, then $\Delta(f(A\mathbf{x})) = (\Delta f)(A\mathbf{x})$.

Hints and comments for Exercise set 6

Exercise 1: The easiest approach is to think about connectedness. For (1), observe that $\mathfrak{g}$ is connected and exp is continuous, so the image $\exp(\mathfrak{g}) \subseteq G$ is connected. For (2), we know that a Lie group satisfying such condition cannot be connected.

Remark. Nevertheless, there are (more complicated) connected examples for (1). One is $\mathsf{SL}(2;\mathbb{R})$, see [Hall, Ch. 3, Exe. 22].

Exercise 2: (1) As seen in class, it is perhaps more convenient to show that the map is an action, it is linear, and continuous (rather than translating everything into square matrices).
(2) There are some natural examples; you need to show that they are the only ones. See also Problem Set 2, Exe. 4.
(3) Keep in mind that an invariant subspace $H \subseteq V$ is itself a representation, and that *its* invariant subspaces are also invariant subspaces of V.
(4) Use differentiation at 0, as seen in class.

Exercise 3: This is a generalization of Problem Set 2, Exe. 4. Guessing the two invariant subspaces H_1, H_2 and proving they are invariant should not be hard. To show that they are the only ones, let H be any invariant subspace. You need to show that if H is not contained in the 1-dimensional invariant subspace H_1, then $H_2 \subseteq H$, and therefore $H = H_2$ or $H = V$. Since H is a subspace, it is closed under linear combinations; since it is invariant, it is closed under permutations of coordinates; use these "moves" to generate H_2 starting from a $\mathbf{v} \in H$.

Remark. In case you are wondering: the presence of f^{-1} in the definition of this action is a technicality, required to guarantee that this map is an action:

$$[g \cdot (f \cdot \mathbf{v})]_i = [f \cdot \mathbf{v}]_{g^{-1}(i)} = \mathbf{v}_{f^{-1}(g^{-1}(i))} = \mathbf{v}_{(gf)^{-1}(i)} = [(gf) \cdot \mathbf{v}]_i \Rightarrow g \cdot (f \cdot \mathbf{v}) = (gf) \cdot \mathbf{v}.$$

The same definition without inverse would not work, since

$$[g \cdot (f \cdot \mathbf{v})]_i = [f \cdot \mathbf{v}]_{g(i)} = \mathbf{v}_{f(g(i))} \neq \mathbf{v}_{(gf)(i)} = [(gf) \cdot \mathbf{v}]_i \Rightarrow g \cdot (f \cdot \mathbf{v}) \neq (gf) \cdot \mathbf{v}.$$

Exercise 4: This is similar to what we did in class for $\mathsf{SO}(2)$.

Exercise 5: An invariant subspace of dimension 1 must be an eigenspace of $\Pi(s)$ for all s.

Bonus exercise: Just apply the chain rule to compute partial derivatives of $f(A\mathbf{x})$. At some point, you will need to use the fact that $A^\intercal A = I$, since $A \in \mathsf{O}(3)$.

Remark. In fact, this calculation proves the claim in any dimension.